

Source Code

```
#include<iostream>
using namespace std;

void marge(int left[], int leftSize, int right[],
int rightSize, int result[]){
    int i = 0, j = 0, k = 0;

    while (i < leftSize && j < rightSize){
        if(left[i] < right[j]){
            result[k] = left[i];
            i++;
        }
        else{
            result[k] = right[j];
            j++;
        }
        k++;
    }

    while(i < leftSize){
        result[k] = left[i];
        i++;
        k++;
    }

    while(j < rightSize){
        result[k] = right[j];
        j++;
        k++;
    }
}

void margeSort(int arr[], int size){
    if(size <= 1){
        return;
    }

    int mid = size / 2;

    int leftHalf[mid];
    int rightHalf[size - mid];

    for(int i = 0; i < mid; i++){
        leftHalf[i] = arr[i];
    }

    for(int i = mid; i < size; i++){
```

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        rightHalf[i - mid] = arr[i];
    }

    margeSort(leftHalf, mid);
    margeSort(rightHalf, size - mid);

    marge(leftHalf, mid, rightHalf, size - mid,
arr);
}

int main(){
    int size;

    cout << "Enter array size: ";
    cin >> size;

    int arr[size];

    cout << "Enter elements: ";

    for(int i = 0; i < size; i++){
        cin >> arr[i];
    }

    margeSort(arr, size);

    cout << "Sorted array: ";

    for(int i = 0; i < size; i++){
        cout << arr[i] << " ";
    }

    return 0;
}
```

Output

```
PS D:\DSA II\Merge Sort> cd "d:\DSA II\Merge Sort\" ;
Enter array size: 7
Enter elements: 38 27 43 3 9 82 10
Sorted array: 3 9 10 27 38 43 82
PS D:\DSA II\Merge Sort> █
```